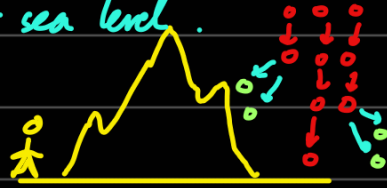


3. Decaying muons: Muons are unstable and decay according to the radioactive decay law $N = N_0 \exp(-0.693 t/t_{1/2})$, where N_0 and N are the number of muons at times $t = 0$ and t , respectively, while $t_{1/2}$ is the half life. The half life of the muons in their own rest frame is 1.52×10^{-6} s. Consider a detector on top of a 2,000 m mountain which counts the number of muons traveling at the speed of $v = 0.98 c$. Over a given period of time, the detector counts 10^3 muons. When the relativistic effects are taken into account, how many muons can be expected to reach the sea level?

Let T : time taken to reach sea level as measured at sea level.

$$\text{Then } T = \frac{H}{v} = \frac{2 \cdot 10^3}{0.98 \cdot 3 \cdot 10^8} = 6.8 \cdot 10^{-6} \text{ s}$$



Now $N = N_0 e^{-t/\tau}$ where $N \rightarrow$ no. of muons reaching sea level.

$t \rightarrow$ time measured in muon's frame to reach sea level.

$$\text{But } t = \frac{T}{\gamma}$$

$$\Rightarrow N = N_0 e^{-\frac{T}{\gamma \tau}} = (1000) \exp\left(-0.693 \cdot \frac{6.8 \cdot 10^{-6}}{1.52 \cdot 10^{-6}} \sqrt{1-0.98^2}\right)$$

$$N \approx 539$$

Q: What if you didn't account for relativity?

A)

$$N = N_0 \exp\left(-\frac{T}{\tau}\right) = (1000) \exp\left(-0.693 \cdot \frac{6.8 \cdot 10^{-6}}{1.52 \cdot 10^{-6}}\right)$$

$$N \approx 45 \quad \times$$

5. Form invariance of the Minkowski line element: Show that the following Minkowski line element is invariant under the Lorentz transformations:

$$ds^2 = c^2 dt^2 - d\mathbf{x}^2.$$

When the boost velocity $\mathbf{v}$ is in an arbitrary vector direction with the boost vector $\boldsymbol{\beta} = \mathbf{v}/c$, then the transformation from an unprimed spacetime coordinate system to a primed coordinate system is given by^[14]

$$\begin{bmatrix} ct' \\ x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \gamma & -\gamma\beta_x & -\gamma\beta_y & -\gamma\beta_z \\ -\gamma\beta_x & 1 + \frac{\gamma^2}{1+\gamma}\beta_x^2 & \frac{\gamma^2}{1+\gamma}\beta_x\beta_y & \frac{\gamma^2}{1+\gamma}\beta_x\beta_z \\ -\gamma\beta_y & \frac{\gamma^2}{1+\gamma}\beta_x\beta_y & 1 + \frac{\gamma^2}{1+\gamma}\beta_y^2 & \frac{\gamma^2}{1+\gamma}\beta_y\beta_z \\ -\gamma\beta_z & \frac{\gamma^2}{1+\gamma}\beta_x\beta_z & \frac{\gamma^2}{1+\gamma}\beta_y\beta_z & 1 + \frac{\gamma^2}{1+\gamma}\beta_z^2 \end{bmatrix} \begin{bmatrix} ct \\ x \\ y \\ z \end{bmatrix},$$

where the **Lorentz factor** is $\gamma = 1/\sqrt{1 - \beta^2}$. The **determinant** of the transformation matrix is +1 and its **trace** is $2(1 + \gamma)$. The inverse of the transformation is given by reversing the sign of $\boldsymbol{\beta}$.

Transformation using rapidity done in class (though that was for only dx not $d\mathbf{u}$)

1. Superluminal motion: Consider a blob of plasma that is moving at a speed v along a direction that makes an angle θ with respect to the line of sight. Show that the *apparent* transverse speed of the source, projected on the sky, will be related to the actual speed v by the relation

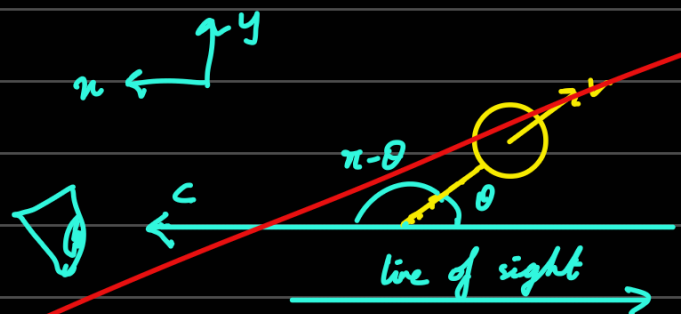
$$v_{\text{app}} = \frac{v \sin \theta}{1 - (v/c) \cos \theta}.$$

From this expression conclude that the apparent speed v_{app} can exceed the speed of light.

From class, $u_y = \frac{u_y'}{\gamma \left(1 + \frac{v u_x'}{c^2}\right)}$

$$u_y = \frac{v \sin(\pi - \theta)}{\gamma \left(1 + \frac{v \cos(\pi - \theta)}{c}\right)}$$

$$\Rightarrow u_y = \frac{v \sin \theta}{\gamma \left(1 - \frac{v \cos \theta}{c}\right)}$$



$$\text{For } u_y \geq c, \quad \frac{v \sin \theta}{1 - \frac{v}{c} \cos \theta} \geq c$$

$$\Rightarrow v \sin \theta \geq c - v \cos \theta$$

$$\Rightarrow \sin \theta + \cos \theta \geq \frac{c}{v}$$

$$\Rightarrow \sin\left(\theta + \frac{\pi}{4}\right) \geq \frac{c}{\sqrt{2}v}$$

$$\Rightarrow \theta \geq \sin^{-1}\left(\frac{c}{\sqrt{2}v}\right) - \frac{\pi}{4} \quad \left(\text{as long as } v \geq \frac{c}{\sqrt{2}}\right)$$

$$\text{So for such } \theta > \sin^{-1}\left(\frac{c}{\sqrt{2}v}\right) - \frac{\pi}{4}, \quad v_{\text{app}} > c$$

Wtf? let's calculate u_x .

$$u_x = \underline{c - v \cos \theta}$$



4. Binding energy: As you may know, the deuteron which is the nucleus of deuterium, an isotope of hydrogen, consists of one proton and one neutron. Given that the mass of a proton and a neutron are $m_p = 1.673 \times 10^{-27}$ kg and $m_n = 1.675 \times 10^{-27}$ kg, while the mass of the deuteron is $m_d = 3.344 \times 10^{-27}$ kg, show that the binding energy of the deuteron is about 2.225 MeV.

Note: MeV refers to Million electron Volts, and an electron Volt is 1.602×10^{-19} J.

$$BE = (m_p + m_n - m_d) c^2$$

$$= 2.225 \text{ MeV}$$

2. Aberration of light: Consider two inertial frames S and S' , with the frame S' moving along the x -axis with a velocity v with respect to the frame S . Let the velocity of a particle in the frames S and S' be $\mathbf{u}$ and $\mathbf{u}'$, and let θ and θ' be the angles subtended by the velocity vectors with respect to the common x -axis, respectively.

(a) Show that

$$\tan \theta = \frac{u' \sin \theta'}{\gamma [u' \cos \theta' + v]},$$

where $\gamma = [1 - (v/c)^2]^{-1/2}$.

(b) For $u = u' = c$, show that

$$\cos \theta = \frac{\cos \theta' + (v/c)}{1 + (v/c) \cos \theta'}$$

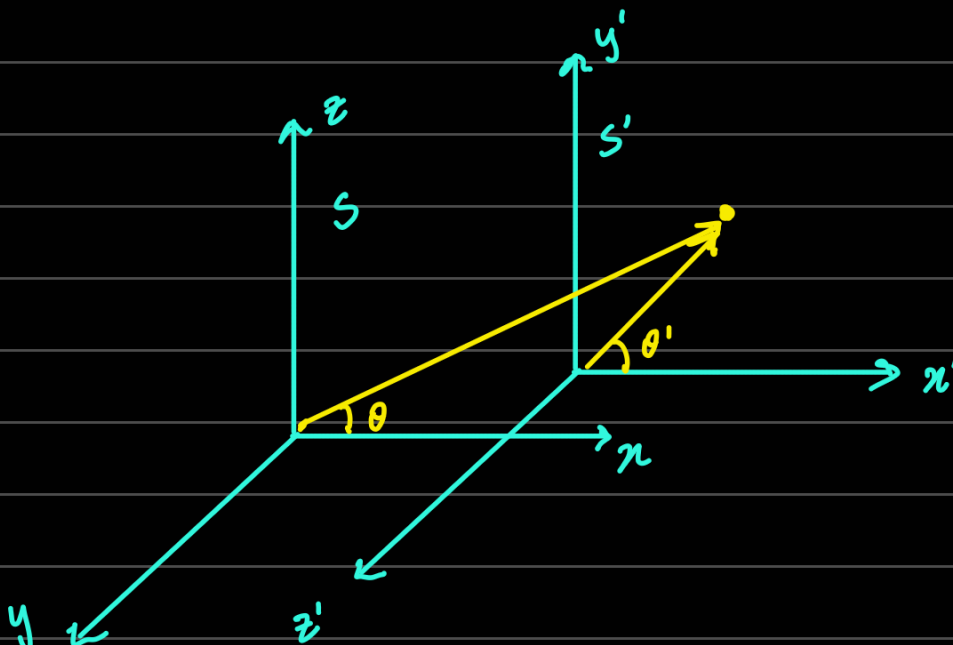
and

$$\sin \theta = \frac{\sin \theta'}{\gamma [1 + (v/c) \cos \theta']}.$$

(c) For $(v/c) \ll 1$, show that

$$\Delta \theta = (v/c) \sin \theta',$$

where $\Delta \theta = (\theta' - \theta)$.



$$a) \tan \theta = \frac{u_y}{u_x} \quad \text{and} \quad u_y = \frac{u_y'}{\gamma \left(1 + \frac{v u_x'}{c^2}\right)}$$

$$u_x = \frac{u_x' + v}{1 + \frac{v u_x'}{c^2}}$$

$$\therefore \tan \theta = \frac{u_y'}{\gamma(u_x' + v)} = \frac{u' \sin \theta'}{\gamma(u' \cos \theta' + v)}$$

$$b) \quad u = u' = c, \quad c \cos \theta = \frac{c \cos \theta' + v}{1 + \frac{v \cos \theta'}{c}}$$

$$\Rightarrow \cos \theta = \frac{\cos \theta' + \beta}{1 + \beta \cos \theta'}$$

$$\beta = \frac{v}{c}$$

$$c \sin \theta = \frac{c \sin \theta'}{\gamma \left(1 + \frac{v \cos \theta'}{c}\right)}$$

$$\Rightarrow \sin \theta = \frac{\sin \theta'}{\gamma (1 + \beta \cos \theta')}$$

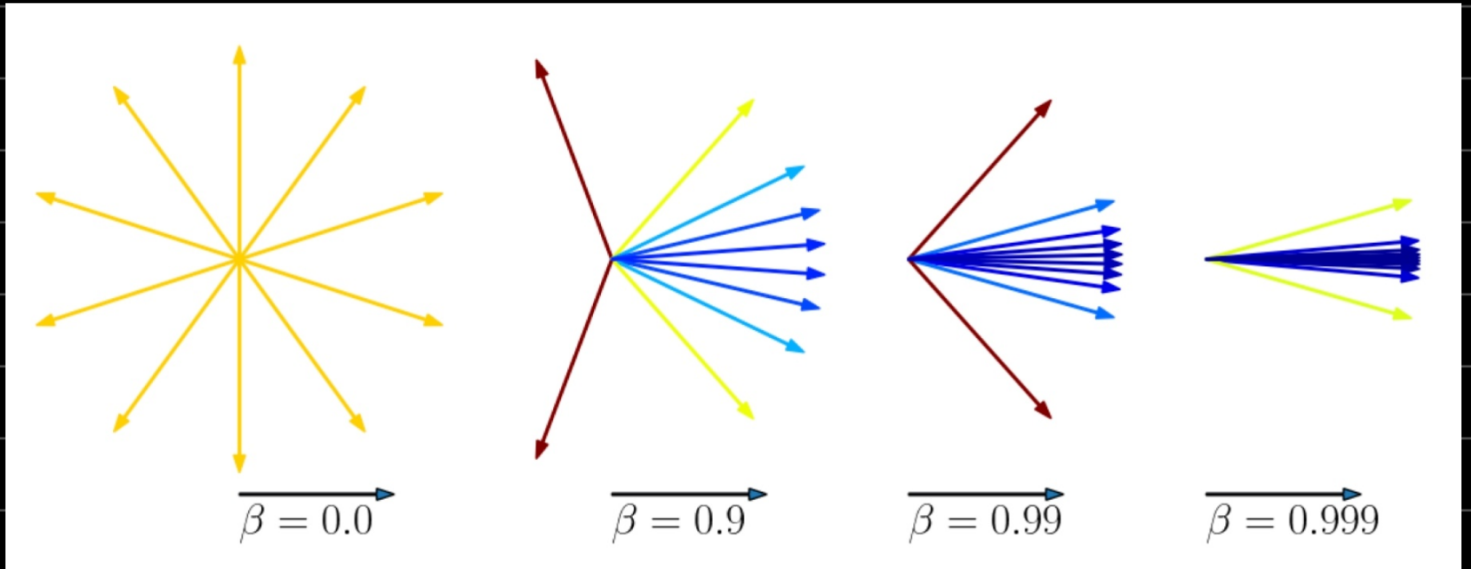
$$c) \quad \sin \theta' - \sin \theta = 2 \sin \left(\frac{\theta' - \theta}{2} \right) \cos \left(\frac{\theta' + \theta}{2} \right)$$

$$\sin \theta' \left(1 - \frac{\sqrt{1 - \beta^2}}{(1 + \beta \cos \theta')} \right) = \Delta \theta \cos \theta' \quad (\theta \approx \theta')$$

$$\sin \theta' \left(1 - \left(1 - \frac{\beta^2}{2} \right) (1 + \beta \cos \theta') \right) = \Delta \theta \cos \theta'$$

$$\sin \theta' \left(1 - \left(1 - \beta \cos \theta' \right) \right) = \Delta \theta \cos \theta'$$

$\Rightarrow$ $\Delta \theta = \beta \sin \theta'$ Difference in angle subtended increases with velocity.



Say a point light source is kept in S' . Then an observer in S sees the above angular distribution.